ECE 340: Semiconductor Devices

Lecture 12: Fermi level at equilibrium & Optical absorption

Prof. Hyunseok Kim

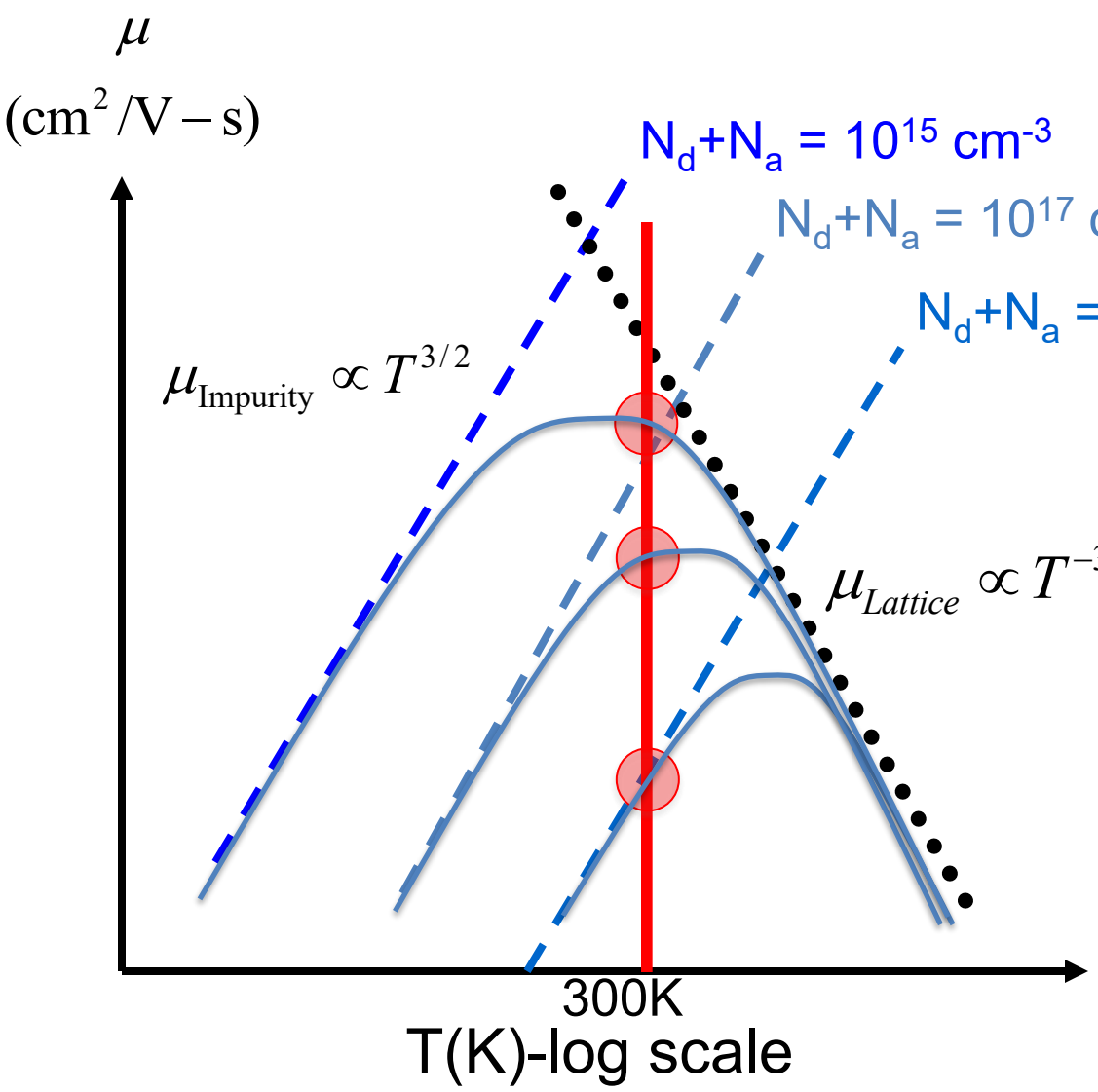
Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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(Recap) Effects of Both Temperature and Impurity Concentration on Mobility

$$\frac{1}{\mu} = \frac{1}{\mu_{Impurity}} + \frac{1}{\mu_{Lattice}} + \frac{1}{\mu_{Defect}} + \ldots$$



At T=300K

- For $N_d+N_a=10^{19}$cm^{-3}, the mobility is dominated by impurity scattering
- For $N_d+N_a=10^{17}$cm^{-3}, the mobility is dominated by impurity scattering
- For $N_d+N_a=10^{15}$cm^{-3}, the mobility is dominated by lattice scattering

(Recap) Hall Voltage and Hall Coefficient

In Steady State, $F_y = 0$

$$E_y = v_x B_z$$

(recall, J=qpv)

$$E_y = \frac{J_x}{qp_o} B_z = R_H J_x B_z$$

Hall coefficient $$R_H = \frac{1}{qp_o}$$

Hall Voltage:

$$V_{AB} = E_y w$$

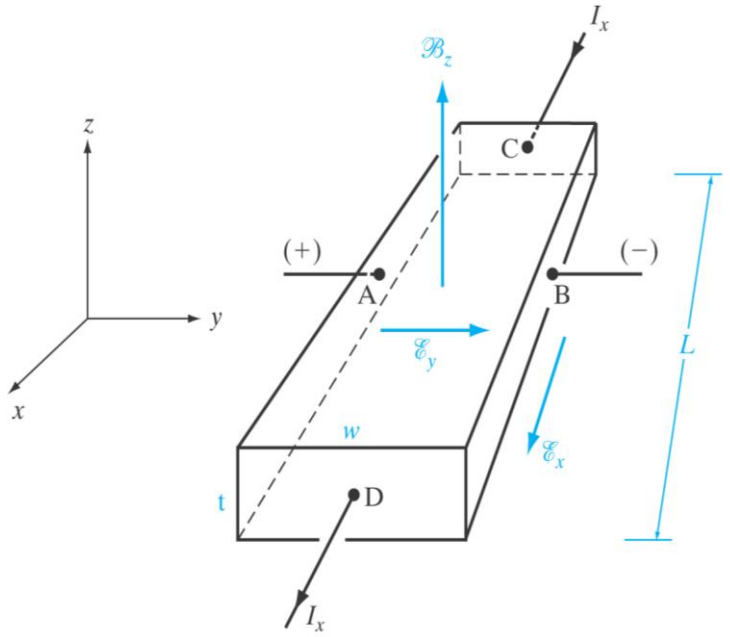


Figure 3.25

The Hall effect.

$$F = q(E + v \times B)$$

$$F_y = q(E_y - v_x B_z)$$

In Steady State: $F_y = 0$

$$E_y = v_x B_z$$

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(Recap) Carrier Concentration and Mobility

Carrier Concentration

- Apply a known magnetic field B_z
- Apply a known current I_x
- Measure the Hall voltage induced across the sample
- Calculate the carrier concentration

$$E_y = v_x B_z = \frac{J_x}{qp_o} B_z$$

$$p_o = \frac{J_x}{qE_y} B_z = \frac{(I_x / wt)B_z}{q(V_{AB} / w)}$$

$$p_o = \frac{I_x B_z}{qtV_{AB}}$$

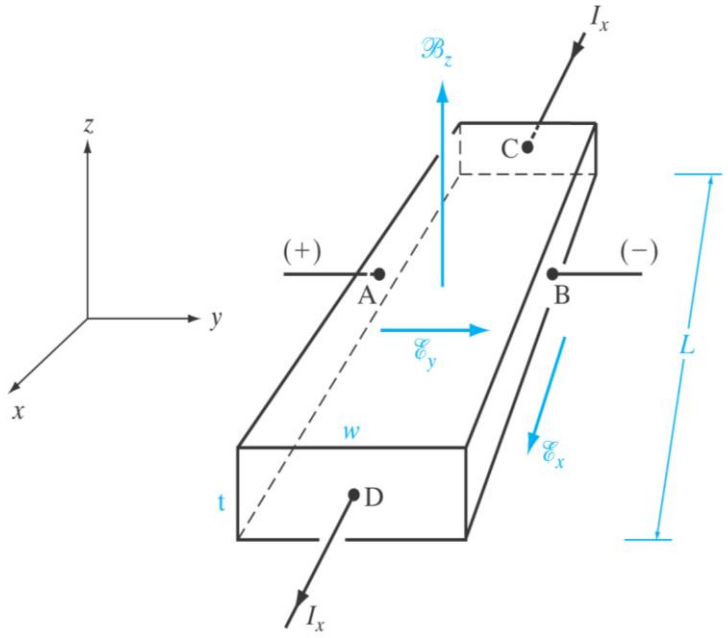


Figure 3.25

The Hall effect.

(Recap) Carrier Concentration and Mobility

Mobility

- Apply a known magnetic field B_z
- Apply a known current I_x
- Measure the Hall voltage induced across the sample
- Measure the resistance across the length of the sample (V_{CD}/I_x)
- Calculate mobility

$$\rho(\Omega - cm) = R\frac{wt}{L} = \frac{V_{CD}}{I_x}\frac{wt}{L}$$

$$\sigma = \frac{1}{\rho} = q\mu_p p_o$$

$$\mu_p = \frac{1}{\rho}\frac{1}{qp_o} = \frac{R_H}{\rho}$$

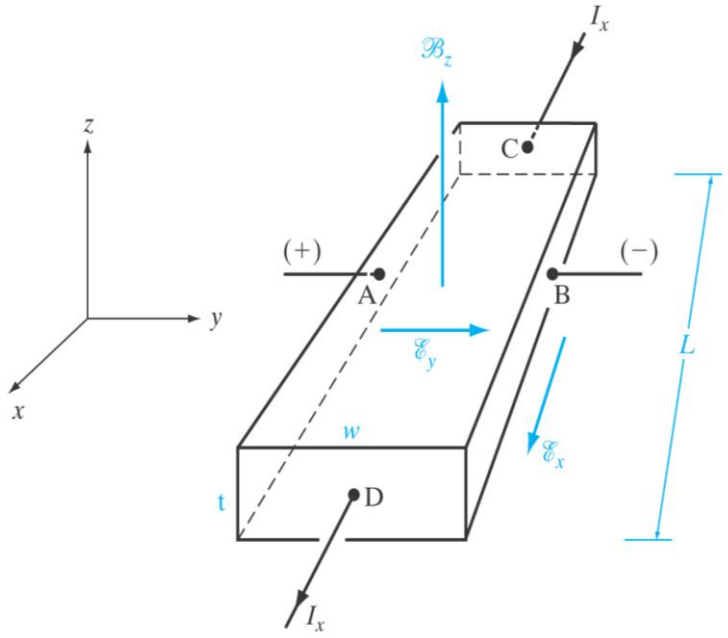
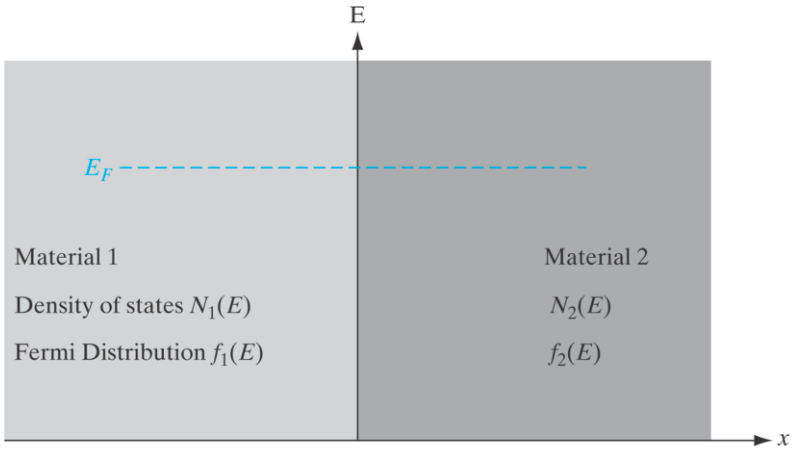


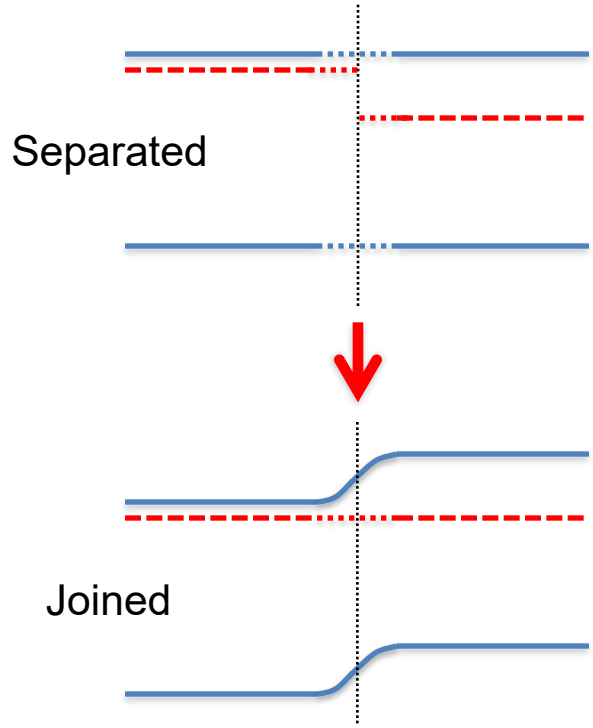
Figure 3.25
The Hall effect.

Invariance of the Fermi Level at Equilibrium

Key Concept

- No Discontinuity or Gradient Can Be Present in the **Equilibrium** Fermi Level E_F
  - In a drawing of energy versus position, E_F is a horizontal line that other energies are referenced against





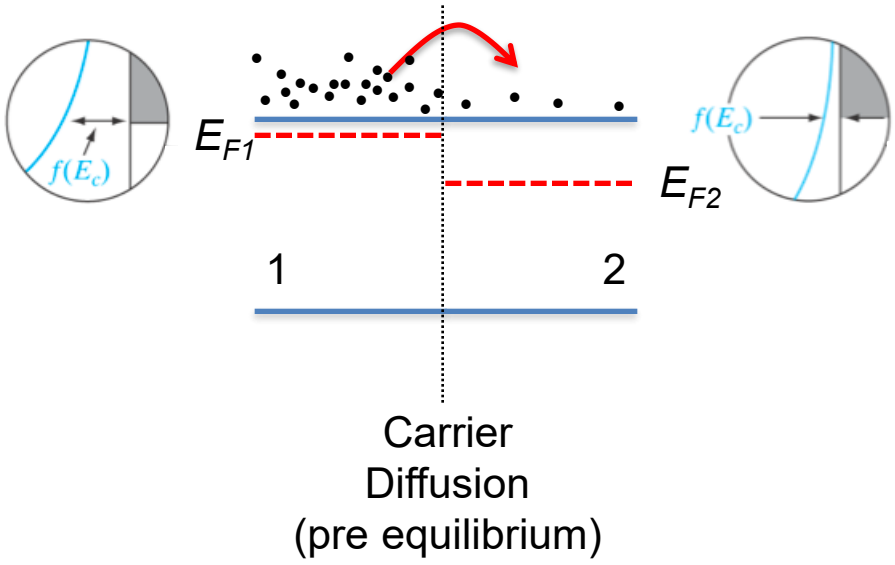
Why?

High Level Explanation

- The Fermi level tells us the probability that a state is occupied at a given energy
- Across a piece of material, a different probability of occupancy at a given energy would lead to a net motion of carriers toward the region with a lower probability
- A net motion of carriers would lead to a net current, which cannot be present in equilibrium
  - Comment: Pay attention to the word “net” – carriers can be in motion due to drift and diffusion as long as the motion is balanced. (This, in fact, is how the Fermi level equilibrates.) More on this later.

Fermi Level Misalignment: Net Flow

- "Across a piece of material, a different probability of occupancy at a given energy would lead to a net motion of carriers toward the region with a lower probability"
- A net motion of carriers is induced.



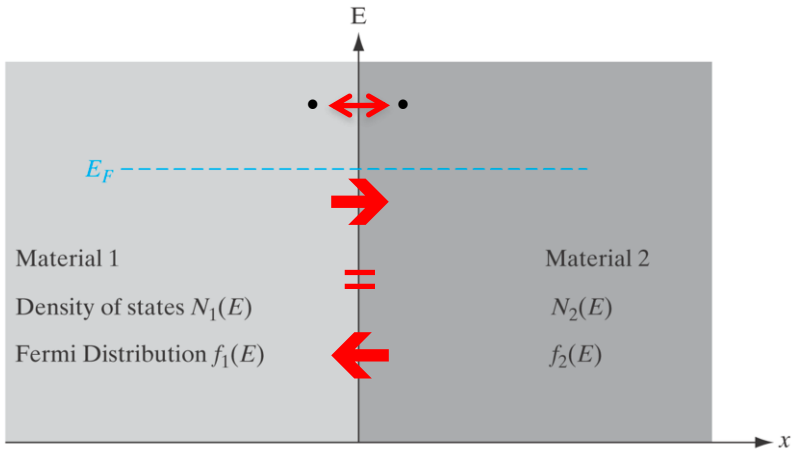
- For any given E:

  $f_1(E) > f_2(E)$ (misaligned)

Another Explanation

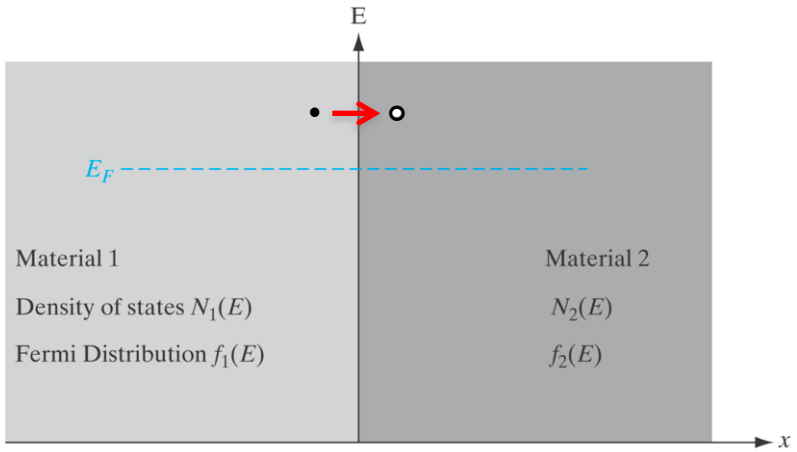
In Equilibrium, there is:

- No net current
- No net charge transport
- For 2 joined materials (1 and 2), the transfer of electrons from material 2 to 1 must exactly balance the transfer from material 1 to 2



Case 1: Electrons Moving From 1 to 2

- At a given energy "E" the motion of an electron from material 1 to material 2 requires a full state in material 1 and an empty state in material 2
- The rate of motion is proportional to the product of the number of electrons at energy "E" in material 1 and the number of empty states at energy "E" in Material 2



Density of States in Material 1: $N_1(E)$

Density of States in Material 2: $N_2(E)$

Probability State Occupied in Material 1: $f_1(E)$

Probability State Empty in Material 2: $\left[1-f_2(E)\right]$

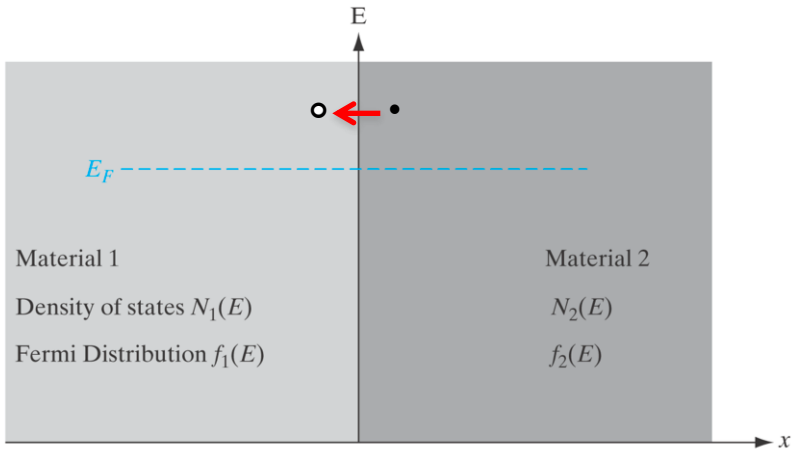
$$\text{Rate}_{1\to 2}: \quad R = a\left\{N_1(E)f_1(E)\right\}\left\{N_2(E)\left[1-f_2(E)\right]\right\}$$

electrons

empty states

Case 2: Electrons Moving From 2 to 1

- At a given energy "E" the motion of an electron from material 2 to material 1 requires a full state in material 2 and an empty state in material 1
- The rate of motion is proportional to the product of the number of electrons at energy "E" in material 2 and the number of empty states at energy "E" in Material 1



Density of States in Material 1: $N_1(E)$

Density of States in Material 2: $N_2(E)$

Probability State Empty in Material 1: $\left[1-f_1(E)\right]$

Probability State Occupied in Material 2: $f_2(E)$

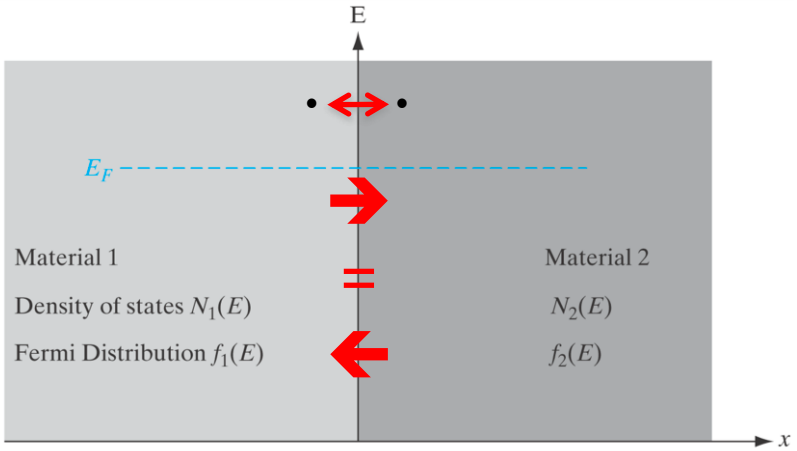
$$\text{Rate}_{2\to1}: \quad R = a\left\{N_1(E)\left[1-f_1(E)\right]\right\}\left\{N_2(E)f_2(E)\right\}$$

empty states

electrons

Equilibrium Fermi Level

- There is no net charge transfer or energy transfer between regions in equilibrium, so the rates from Case 1 and Case 2 must balance
  - We assume that the constant of proportionality "a" for the rate of motion from 1 to 2 is the same as that from 2 to 1 (reciprocity)



$$R_{1\to 2} = aN_1(E)f_1(E)\times N_2(E)\left[1-f_2(E)\right] = aN_2(E)f_2(E)\times N_1(E)\left[1-f_1(E)\right] = R_{2\to 1}$$

so

$$N_1(E)f_1(E)N_2(E) - N_1(E)f_1(E)N_2(E)f_2(E) = N_2(E)f_2(E)N_1(E) - N_2(E)f_2(E)N_1(E)f_1(E)$$

multiplying through by $\dfrac{1}{N_1(E)N_2(E)}$:

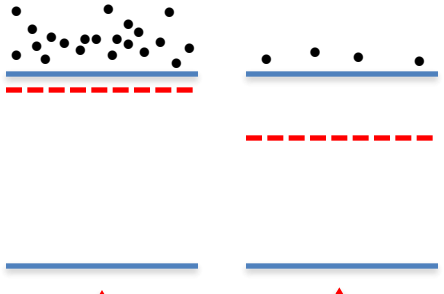
$$f_1(E) - f_1(E)f_2(E) = f_2(E) - f_2(E)f_1(E) \quad \text{therefore}$$

$$f_1(E) = f_2(E) \quad \text{and}$$

$$\frac{1}{1+e^{(E-E_{F1})/kT}} = \frac{1}{1+e^{(E-E_{F2})/kT}} \quad \Rightarrow \quad E_{F1} = E_{F2}$$

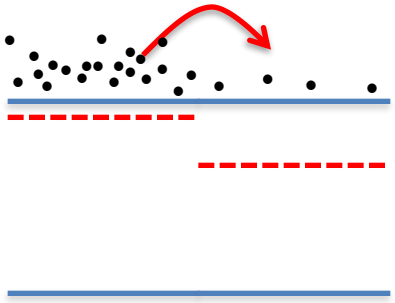
$$\boxed{\frac{dE_F}{dx} = 0}$$

Establishing Equilibrium

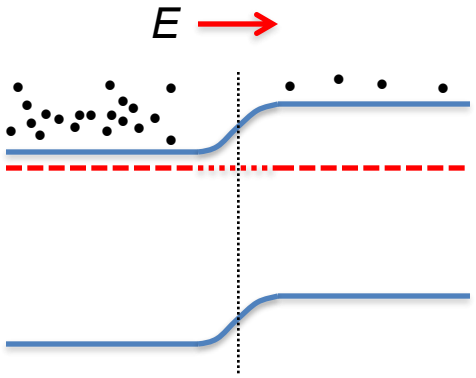


High Carrier Concentration

Low Carrier Concentration



Carrier Diffusion (pre equilibrium)

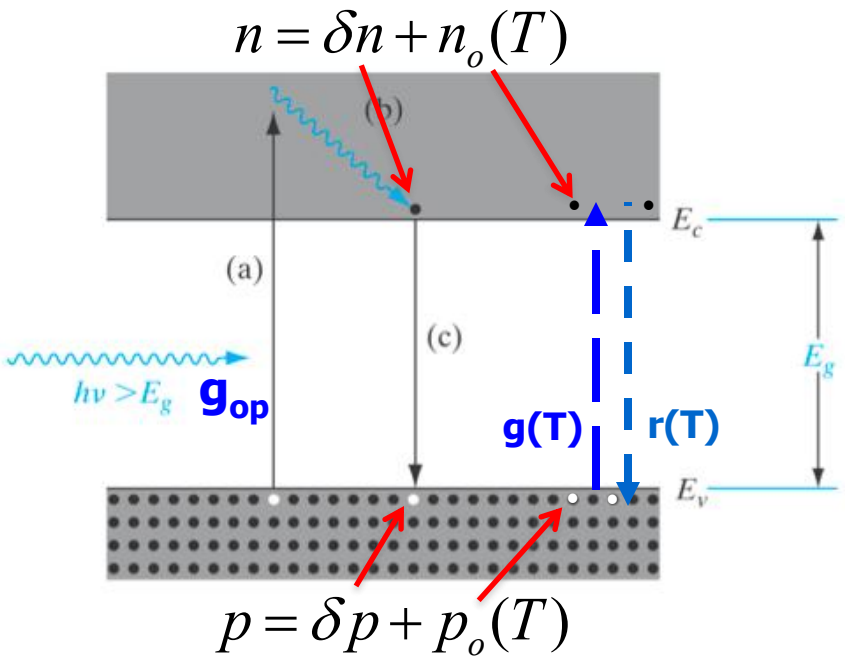


Equilibrium: Diffusion Balanced by Drift (Electric Field)

We Will Come Back to this in the Discussion of pn Junctions!

Optical Absorption

Optical Absorption of Photons and Excess Carriers



- A photon with energy $hv<E_g$ will not be absorbed. Hence, it is transmitted through the sample.
  - Insulators are frequently transparent due to their large bandgap
- A photon with energy $hv>E_g$ can be absorbed to excite an electron from the valance band to the conduction band.
- Thermal generation of carriers provides the equilibrium carrier concentrations, $n_o(T)$ and $p_o(T)$
- Optical generation of excess carriers (δn and δp) can contribute to the conductivity

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Terminology

- ***Excess Carriers:*** Concentration of carriers above the equilibrium concentration created by some perturbing event (optical injection, current injection, etc.)
  - For optical absorption, excess carriers are created in pairs (1 electron + 1 hole)
- ***Absorption Coefficient "α":*** how rapidly light intensity decreases as it travels through a material due to absorption

Total Absorption

I_0 : incident photon flux

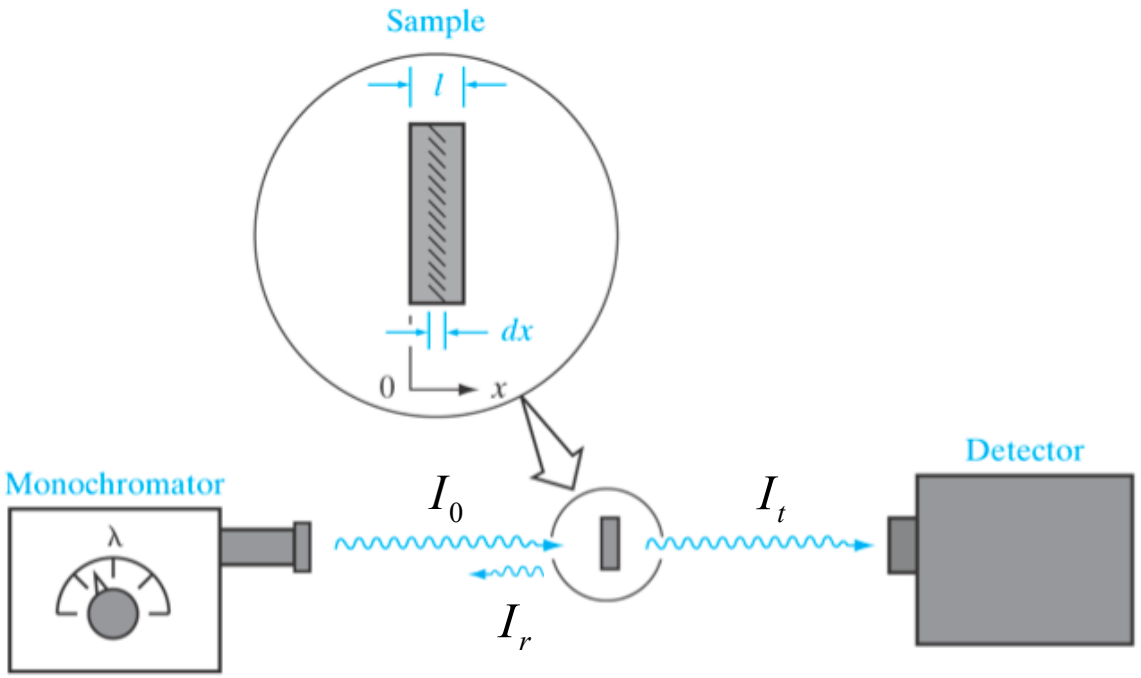
I_t : transmitted photon flux

I_r : reflected photon flux

I_a : photons absorbed per unit area per unit time

$I(x)$: photon flux at "x"

All have units of $\frac{\text{photons}}{\text{cm}^2 \cdot \text{s}}$

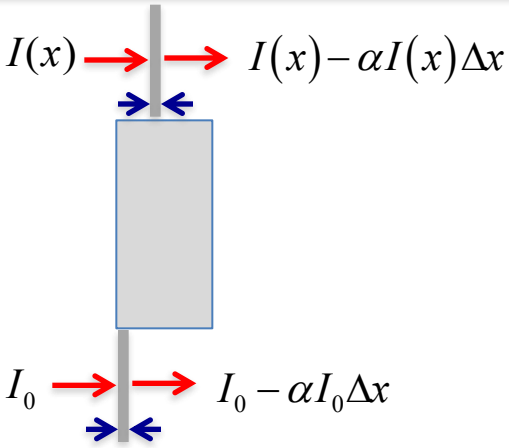


Incident photons are either absorbed, transmitted, or reflected:

$I_o = I_a + I_t + I_r$

Note: Streetman assumes $I_r \sim 0$

What Happens Inside the Sample?



thickness
Δx

$$I(x+\Delta x)=I(x)-\alpha I(x)\Delta x$$

$$I(x+\Delta x)-I(x)=-\alpha I(x)\Delta x$$

$$\frac{I(x+\Delta x)-I(x)}{\Delta x}=-\alpha I(x)$$

$$\lim_{\Delta x\to 0}\left(\frac{I(x+\Delta x)-I(x)}{\Delta x}\right)=\frac{dI(x)}{dx}=-\alpha I(x)$$

Differential Equation:

$$-\frac{dI(x)}{dx}=\alpha I(x)$$

Solution:

$$I(x)=\left(I_o-I_r\right)e^{-\alpha x}\sim I_0 e^{-\alpha x}$$

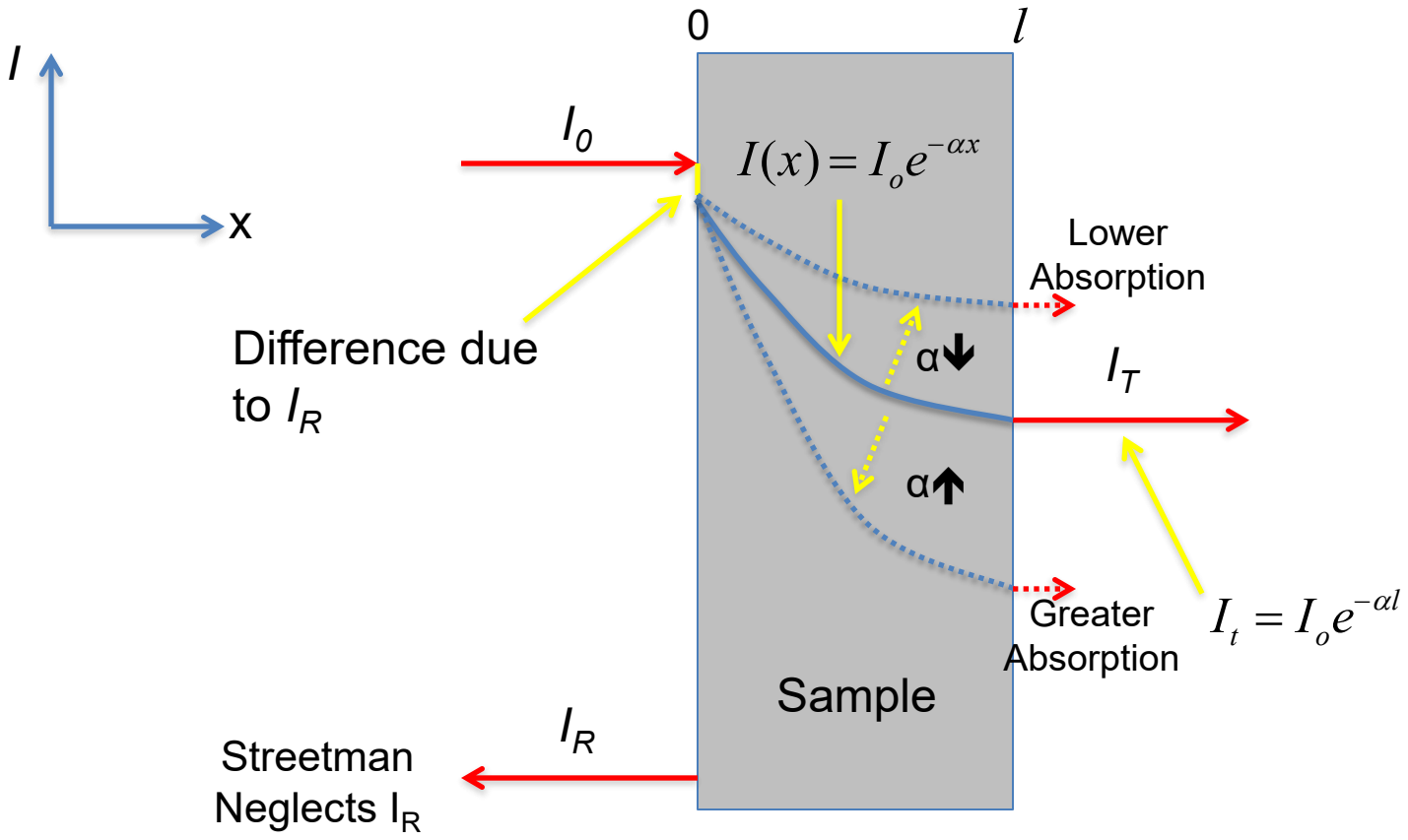
Absorption for sample with thickness "l":

$$I_t\simeq I_o e^{-\alpha l}$$

Note: We've assumed I_R is negligible in this analysis

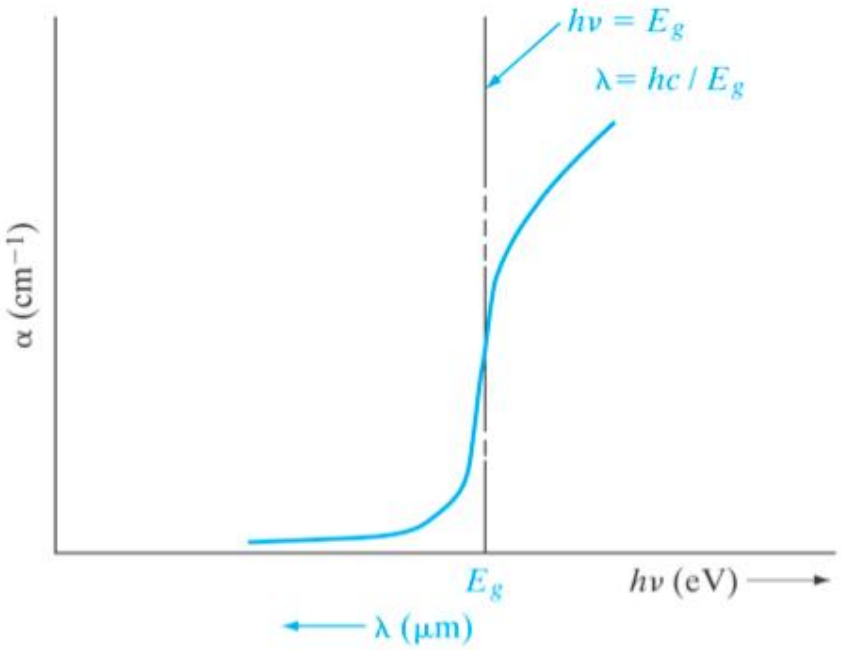
Photon Flux Versus Position

- The photon flux drops exponentially in the sample
- The relationship for the generated carriers is more complicated due to diffusion



Absorption Coefficient

- The absorption coefficient depends upon the incident wavelength λ
- Below the bandgap there is minimal absorption
- The units of absorption are typically cm^{-1}



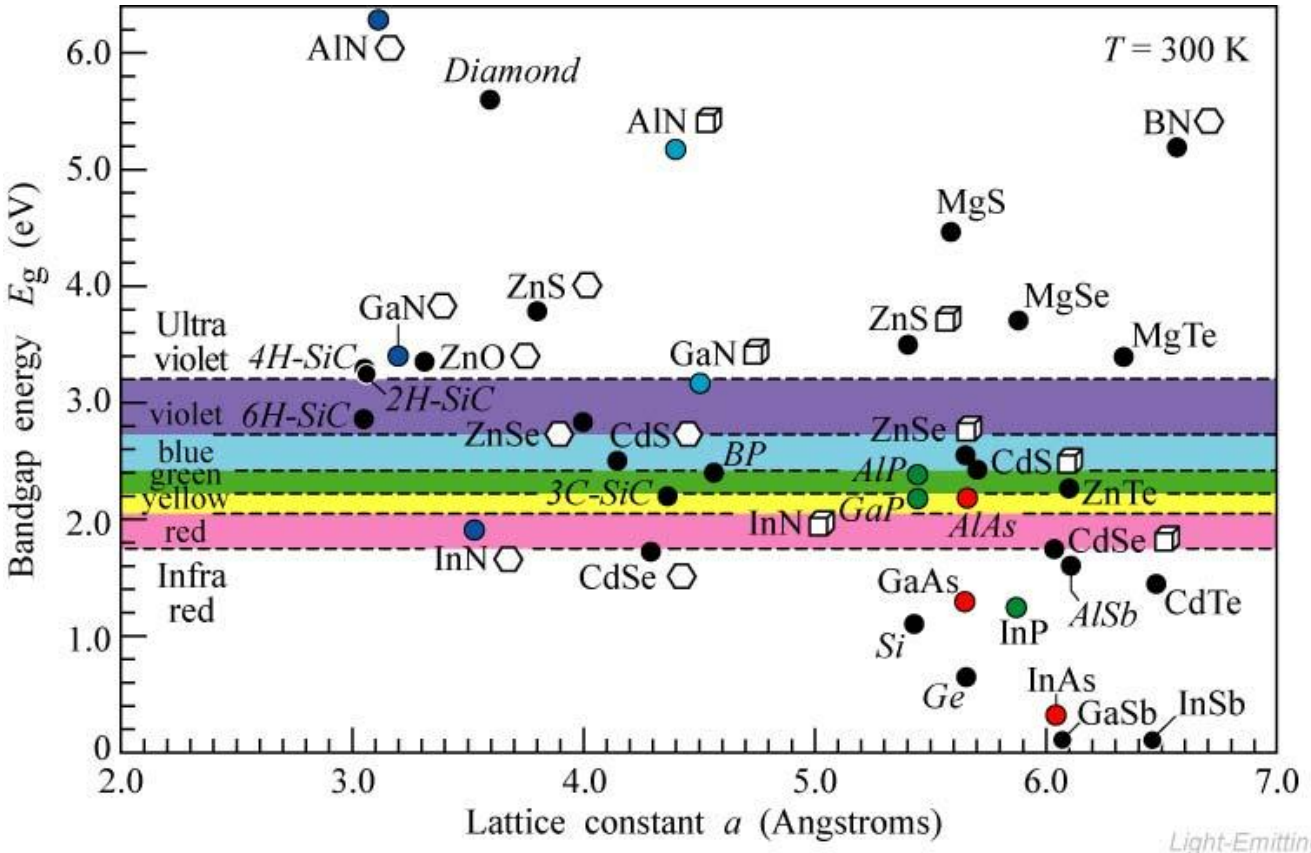
Useful Relationship:

$$E(eV) = h\nu = h\frac{c}{\lambda} = \frac{1.239}{\lambda(\mu m)}$$

A photon with a wavelength of 1 μm has an energy of 1.239 eV

Comments on Optical Absorption

- Semiconductors absorb energies greater than their bandgap, which means wavelengths shorter than the wavelength at their bandgap
- Si, Ge, and GaAs absorbs infrared (and higher energy / shorter λ)
- GaN pushes into the UV (and higher energy / shorter λ)



"*Italics*" = indirect gap

"Roman" = direct gap

⬡ hexagonal structure

cubic structure

Fig. 21.4. Room-temperature bandgap energy versus lattice constant of common elemental and binary compound semiconductors.

E. F. Schubert
Light-Emitting Diodes (Cambridge Univ. Press)
www.LightEmittingDiodes.org

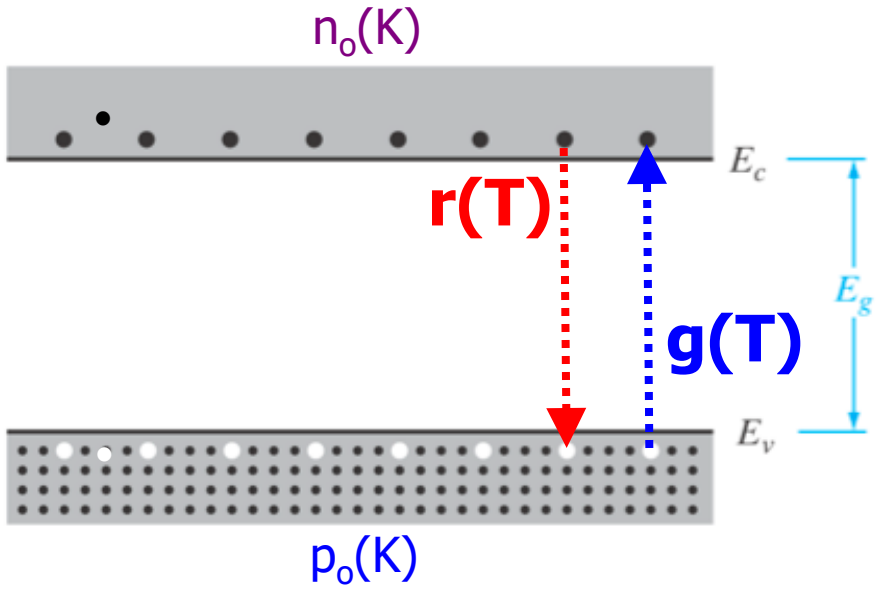
Generation

Terminology Reminder

- ***Equilibrium:*** A special case of steady state. No external perturbing excitation is present except temperature, and there is no net motion of charge.
- ***Steady State***: A condition in which all perturbing processes are constant. In steady state, opposing processes are present that balance perturbing processes.

Equilibrium Generation

Some electrons have sufficient energy to jump to the conduction band (phonon interactions)



Generation and recombination are balanced:

$$g(T) = g_i = r_i = \alpha_r n_o p_o = \alpha_r n_i^2$$

α_r : e-h recombination coefficient
(Not the same as the absorption coefficient α)